

SYNTHESIS OF THYROID HORMONE

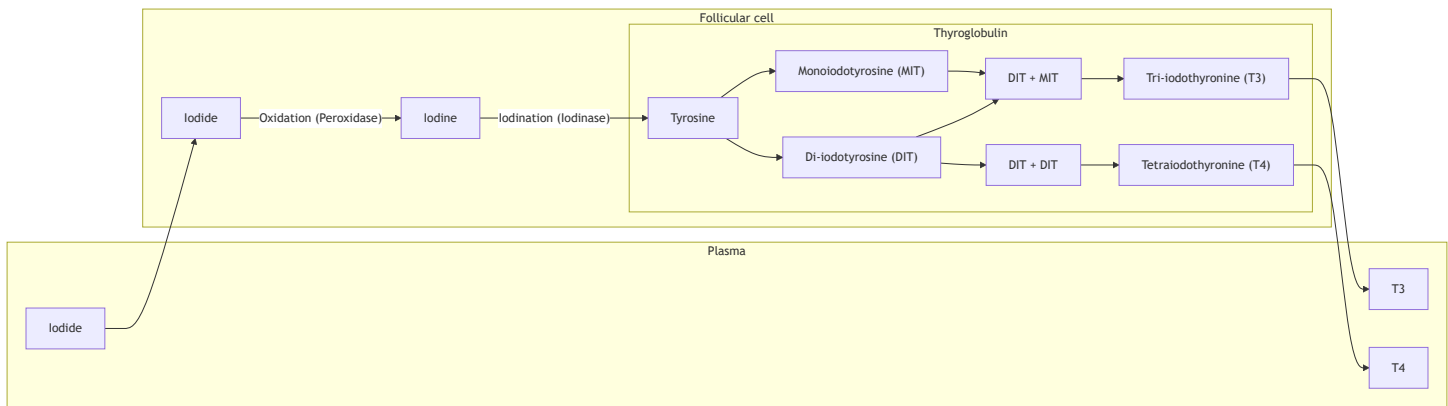
- Takes place in **thyroglobulin**, present in **follicular cavity**.
- **Essential components** :—
 - **Iodine** → *Converted into iodide* → *Absorbed from GI tract*
 - **Tyrosine** (*also absorbed from GI tract*)

STAGES OF SYNTHESIS OF THYROID HORMONE

Occurs in 6 stages :—

1. Thyroglobulin synthesis
2. Iodide trapping
3. Oxidation of iodide
4. Transport of iodine into follicular cavity
5. Iodination of tyrosine
6. Coupling reactions

SYNTHESIS OF THYROID HORMONES (PATHWAY)



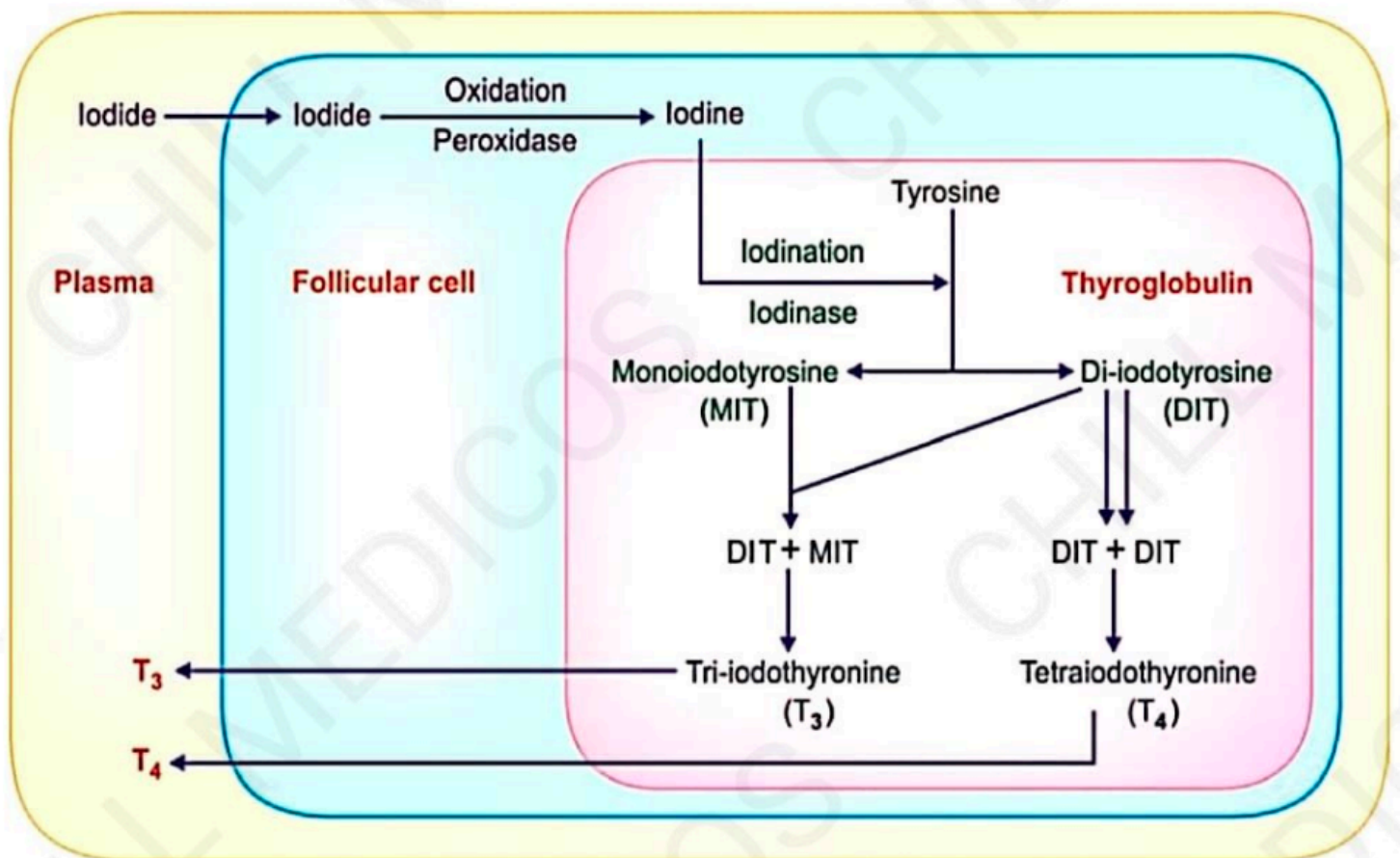


FIGURE 67.3: Synthesis of thyroid hormones

1. THYROGLOBULIN SYNTHESIS

- **ER and Golgi apparatus** in follicular cells of thyroid gland synthesize and secrete thyroglobulin continuously:
- ↓
- **Large glycoprotein** (140 molecules of AA tyrosine)

2. IODINE TRAPPING

- **Iodide actively transported** from blood into follicular cell, against electrochemical gradient.
- **Iodide pump** → Iodide along with Na^+ transported into follicular cell by $\text{Na}^+ - \text{I}^-$ **symport pump**.
- **Normally:** Iodide is 300 times greater concentrated in thyroid gland.
(But during hyperactivity of thyroid gland, it ↑ses 200 times more).

3. OXIDATION OF IODIDE

- **Iodide must be oxidized to iodine:**
 - ↓
 - Because only iodine is capable of combining with tyrosine to form thyroid hormones.
- **Occurs inside follicular cells** in presence of **thyroid peroxidase**.

4. TRANSPORT OF IODINE INTO FOLLICULAR CAVITY

- Transported by an iodide–chloride pump called **pendrin**.

5. IODINATION OF TYROSINE

- **Combination of iodine with tyrosine:**
 - Takes place in thyroglobulin.
- **Iodine binds with thyroglobulin** (*process called **organification of thyroglobulin***):
 - ↓
 - Then, **iodine (I)** combines with **tyrosine**
 - ↓
 - Accelerated by **enzyme iodinase** (*secreted by follicular cells*).

6. COUPLING REACTIONS

- **Iodotyrosine residues (MIT & DIT) get coupled with one another:**

Tyrosine + I = Monoiodotyrosine (MIT)

MIT + I = Di-iodotyrosine (DIT)

DIT + MIT = Tri-iodothyronine (T₃)

MIT + DIT = Reverse T₃

DIT + DIT = Tetraiodothyronine OR Thyroxine (T₄)